1 Introduction and main results

1.1 The SYK model

The Sachdev–Ye–Kitaev (SYK) model is a quantum-mechanical model of N fermionic degrees of freedom with random, all-to-all interactions. It originates from the work of Sachdev and Ye on random quantum magnets in 1993. A related Majorana-fermion model was introduced by Kitaev in 2015 in the context of quantum holography and is now known as the SYK model. Subsequent work, notably by Maldacena and Stanford, clarified its large-N structure and its connections to quantum chaos and nearly two-dimensional gravity. The SYK model is strongly interacting, yet it remains analytically tractable in the large-N limit. Depending on how the inverse temperature $\beta = \beta_N$ scales as $N \to \infty$, the model exhibits several distinct asymptotic regimes. It is quite rare for a strongly interacting quantum system to exhibit such a rich hierarchy of asymptotic regimes while still remaining analytically accessible. This combination makes the SYK model an important testing ground for both mathematics and physics. In particular, many physicists study these regimes in order to understand more precisely how strongly interacting quantum systems behave across different scales, and to identify which mathematical structures and physical mechanisms persist in more realistic models. In this sense, the SYK model serves as a relatively simple prototype from which one can develop tools and ideas for the study of more complicated quantum many-body systems, disordered systems, quantum chaos, and models of quantum gravity.

(i) Fixed-temperature large-N regime (β : fixed): This is the basic mathematical large-N limit of the SYK model. It suppresses finite-size and disorder fluctuations and provides a tractable framework for studying finite-temperature observables and the emergence of the lower-temperature physical regimes. (ii) Conformal scaling regime ($1 \ll \beta \ll N$): The system is strongly interacting and approximately conformal, while finite-N effects remain small. In this regime, the SYK model realizes a non-Fermi liquid, in which conventional quasiparticle excitations are absent and the low-energy behavior is governed instead by collective scale-invariant dynamics. (iv) Quantum-Schwarzian regime ($N \ll \beta$): Quantum fluctuations of the Schwarzian mode become strong, and finite-N effects are essential. In this regime, the SYK model probes the fully quantum dynamics of the same Schwarzian degree of freedom that appears in nearly-AdS_2 gravity. (vi) Finite-size / discrete-spectrum regime ($\liminf_N \log(\beta_N)/N > 0$): Individual energy levels become relevant, and hence, Random matrix theory are applied.

Fix an even integer $q \geq 4$, and let N be even. Write

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}},$$

and let $G = (g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \ldots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I = \mathrm{i}^{q/2} \psi_{i_1} \cdots \psi_{i_q}.$$

Then ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Define

$$H_N(G) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; G) = \operatorname{tr} e^{-zH_N(G)}.$$

Here $\operatorname{tr} = 2^{-N/2} \operatorname{Tr}$ denotes the normalized trace; in particular, $Z_N(0; G) = 1$. This normalization does not affect the relative fluctuations studied below. We sometime write $Z_N(\beta) = Z_N(\beta; G)$ for brevity. For real β, set

$$Y_N(\beta; G) := \frac{Z_N(\beta; G)}{\mathbb{E} Z_N(\beta)}.$$

Thus $\mathbb{E} Y_N(\beta) = 1$. Throughout, all limits are taken as $N \to \infty$ through even integers.

1.2 Main results

It is studied the limiting behavior of the logarithmic partition function in high-temperature regime:

Theorem 1.1. *There exists $\tilde{\beta}_q > 0$ such that for any $\beta \in [-\tilde{\beta}_q, \tilde{\beta}_q]$, there exists $\ell_q(\beta)$ such that*

$$\lim_{N\to\infty} N^{-1} \log Z_N = \lim_{N\to\infty} N^{-1} \log \mathbb{E}[Z_N] = \ell_p(\beta) \qquad a.s.$$

Moreover, $\ell_p(\beta)$ is differentiable over $(-\tilde{\beta}_q, \tilde{\beta}_q)$.

Our main result is the following fluctuation resutls:

Theorem 1.2. *There exists $\beta_q \in (0, \tilde{\beta}_q]$ such that for any $\beta \in (-\beta_q, \beta_q)$,*

$$N^{q/2-1} \left(Y_N(\beta) - 1\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\beta^2(\ell_q'(\beta))^2\right). \tag{1}$$

Moreover,

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{2}$$

Corollary 1.3. *With the same β_p as in Theorem 1.2, for any $\beta \in (-\beta_q, \beta_q)$,*

$$N^{q/2-1} \left(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\,\beta^2(\ell_q'(\beta))^2\right).$$

1.3 Previous work

Rigorous results have recently identified the high-temperature annealed and quenched free-energy limits [**?**]. The purpose of this paper is to determine the leading distributional fluctuations in the same high-temperature regime.

Feng, Tian, and Wei [**?**] proved a central limit theorem for linear eigenvalue statistics of the Gaussian SYK Hamiltonian. Their theorem treats a fixed test function f under regularity assumptions on f and f'. In the normalization used in that paper, however, the partition function corresponds to

$$f_{N,\beta}(x) = e^{-\beta\tau_N x}, \qquad \tau_N \asymp \sqrt{N},$$

so both the test function and its derivative vary with N. Hence, their fixed-test-function theorem therefore does not apply directly to $Z_N(\beta)$.

1.4 Outline of the proof

Recall that $\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)}$. By applying the chaos expansion, since Clifford sign-gauge symmetry eliminates the first Gaussian chaos and identifies the second chaos exactly, we have the following:

$$Y_N(\beta) - 1 = A_N(\beta)Q_N + \mathcal{R}_N(\beta),$$

where

$$A_N(\beta) = \frac{\beta}{2}\partial_\beta \log \mathbb{E}Z_N(\beta), \qquad Q_N = \frac{1}{|\mathcal{I}_N|}\sum_I (g_I^2 - 1). \tag{3}$$

Here $\mathcal{R}_N$ is the sum of the Gaussian chaoses of Y_N of orders at least three. Thus the leading random term is a normalized sum of $\mathcal{I}_N$ independent centered variables. The main task is to prove

$$\|\mathcal{R}_N(\beta)\|_2^2 = o(N^{-(q-2)}).$$

Indeed, a Duhamel derivative bound and the sign-gauge parity rule give

$$\sum_{r\geq 3}\|P_r Y_N(\beta)\|_2^2 \leq \sum_{r\geq 3}(\beta\sigma_N)^{2r}\sum_{\substack{|\alpha|=r\\ \alpha \text{ even}}}\frac{1}{\alpha!}.$$

Using a combinatorial argument, we have that the entire higher-order sum is bounded by $O(N^{-3(q/2-1)}) = o(N^{-(q-2)})$.

Moreover, the annealed free-energy input and convexity imply

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell_q'(\beta).$$

Since $M_N \sim N^q/q!$, the classical central limit theorem for Q_N, the remainder estimate gives the claimed limit for Y_N.